

Independent Verification of Public Road Infrastructure Delivery Using Multi-Temporal Satellite Imagery

Project Proposal and Machine Learning System Design Document

Author

[Your Name]

Duration

2 Weeks

Domain

Remote Sensing, Computer Vision, Public Policy, Government Accountability, Geospatial Machine Learning

1. Executive Summary

Governments worldwide invest billions of dollars annually in public infrastructure projects. Despite transparency initiatives and public procurement systems, independent verification of whether projects are physically completed remains difficult, expensive, and infrequent.

This project proposes an end-to-end machine learning system that uses multi-temporal satellite imagery to independently verify the physical completion status of publicly funded road infrastructure projects in the Philippines.

Rather than attempting to directly detect corruption, the system focuses on a more measurable and defensible objective:

Can satellite imagery identify road projects whose reported completion status is inconsistent with observable physical progress?

The proposed system combines government procurement records, project metadata, and Sentinel-2 satellite imagery to generate project-level completion likelihood scores. Projects exhibiting discrepancies between reported and observed completion may then be prioritized for human review by auditors, journalists, civil society organizations, or government oversight bodies.

The project serves as an early-warning and accountability tool rather than a replacement for formal investigation.

2. Motivation

Infrastructure spending constitutes one of the largest categories of public expenditure.

However, several challenges exist:

- Limited capacity for field inspections
- Geographic dispersion of projects
- Delays in reporting
- Incomplete monitoring coverage
- Potential misalignment between reported and actual progress

Recent advances in satellite imagery and deep learning provide an opportunity to independently assess physical changes associated with infrastructure construction.

The Philippines presents a particularly compelling case because:

- Large public investment in road infrastructure
- Publicly available procurement and project reporting systems
- Frequent natural and logistical challenges affecting project delivery
- Strong societal interest in transparency and accountability

The ability to automatically identify potentially incomplete projects may support:

- Government auditing
 - Public accountability
 - Civil society monitoring
 - Infrastructure planning
 - Evidence-based policymaking
-

3. Research Question

Primary Research Question

Can multi-temporal satellite imagery be used to independently verify whether government-funded road infrastructure projects reported as completed exhibit observable physical completion?

Secondary Research Questions

1. Can a machine learning model distinguish completed and incomplete road projects using satellite imagery alone?

2. Does combining procurement metadata with imagery improve predictive performance?
 3. Can the system generate reliable project-level risk scores suitable for human review?
-

4. Problem Formulation

The project is formulated as a binary classification task.

Given:

- Project metadata
- Project location
- Satellite imagery before construction
- Satellite imagery near completion date
- Satellite imagery after completion date

Predict:

$Y \in \{0,1\}$

Where:

0 = Visibly incomplete or delayed

1 = Visibly completed

Output:

$P(\text{Completed})$

The final risk score is:

$\text{Risk} = 1 - P(\text{Completed})$

Projects with high risk scores become candidates for human review.

5. Scope

Included

- National road projects
- Road widening

- Road rehabilitation
- New road construction
- Philippines only
- Sentinel-2 imagery
- Project-level classification

Excluded

- Bridges
- Railways
- Airports
- Corruption prediction
- Real-time deployment
- Nationwide production systems

The focus on roads is motivated by their larger spatial footprint and visibility in medium-resolution satellite imagery.

6. Related Literature

Infrastructure Monitoring from Satellite Imagery

Recent work has demonstrated that satellite imagery can accurately predict infrastructure quality indicators.

Oshri et al. showed that deep learning models trained on Landsat and Sentinel imagery achieved strong performance in predicting infrastructure quality across Africa.

These findings suggest that physical infrastructure characteristics are observable from remotely sensed imagery.

Road Detection and Mapping

Deep convolutional networks have achieved strong performance in extracting road networks from satellite imagery.

Road construction and road expansion introduce measurable spatial patterns including:

- Vegetation removal
- Surface reflectance changes
- Earthworks
- Linear network expansion

These patterns motivate the use of image-based completion assessment.

Change Detection

Change detection represents one of the most active areas of remote sensing research.

Common approaches include:

- Image differencing
- Siamese CNNs
- Temporal transformers
- Self-supervised representation learning

The proposed system leverages change detection principles by comparing imagery acquired before and after project completion deadlines.

Public Procurement Analytics

Most corruption and procurement risk research relies on:

- Contract metadata
- Network analysis
- Red flag indicators

Few studies integrate physical verification from satellite imagery.

This project bridges procurement analytics and remote sensing.

7. Dataset Construction

Data Source 1: Government Project Records

Potential sources include:

- DPWH Transparency Portal
- Government procurement databases
- Infrastructure reporting portals

Required fields:

Field	Description
Project ID	Unique identifier
Project Name	Infrastructure project
Province	Location
Municipality	Location
Contract Amount	Budget
Award Date	Start date
Completion Date	Planned finish
Reported Status	Completion status

Data Source 2: Satellite Imagery

Source:

Sentinel-2

Characteristics:

- 10-meter resolution
- 5-day revisit frequency
- Free access
- Global coverage

Acquisition windows:

T0 = Before construction

T1 = Midpoint

T2 = Completion date

T3 = 3–6 months after completion

Data Source 3: Road Geometry

OpenStreetMap

Purpose:

- Road alignment verification
 - Spatial context
 - Project boundary estimation
-

8. Label Construction

Completed

Conditions:

Reported completed

AND

Visual inspection confirms road construction or improvement.

Incomplete

Conditions:

Reported delayed

OR

Reported completed but visible progress appears absent.

Annotation Process

A sample of projects will be manually reviewed.

Reviewers inspect:

- Before imagery
- After imagery

and assign labels.

This creates a gold-standard evaluation set.

9. Data Pipeline

Step 1

Acquire project records.

Step 2

Geocode project locations.

Step 3

Create project spatial buffers.

Example:

500m–1000m around project centerline.

Step 4

Download Sentinel imagery.

Step 5

Cloud masking.

Step 6

Temporal alignment.

Step 7

Image chip extraction.

Step 8

Dataset generation.

Output:

(Project Metadata, Image Sequence, Label)

10. Feature Engineering Baseline

Before deep learning, a strong baseline will be developed.

Features:

Spectral

- NDVI
- NDBI
- Surface reflectance statistics

Temporal

- Mean change
- Maximum change
- Trend slope

Spatial

- Texture measures
- Edge density

Model:

XGBoost

Purpose:

Interpretability and benchmarking.

11. Deep Learning System

Architecture

Siamese Convolutional Neural Network

Input A:

Pre-construction image

Input B:

Post-completion image

Shared encoder:

ResNet-18

Feature fusion:

Concatenation

Classification head:

Fully connected network

Output:

$P(\text{Completed})$

Why Siamese Networks?

They are specifically designed for:

- Image comparison
- Change detection
- Limited datasets

and are computationally feasible within project constraints.

12. Training Strategy

Training split:

70% train

15% validation

15% test

Data augmentation:

- Rotation
- Horizontal flips

- Brightness variation

Loss:

Binary Cross Entropy

Optimizer:

Adam

Learning Rate:

1e-4

Early stopping:

Validation loss monitoring

13. Evaluation Framework

Classification Metrics

Accuracy

Precision

Recall

F1 Score

ROC-AUC

PR-AUC

Ranking Metrics

Precision@K

Where:

K = top projects flagged for review.

This metric aligns with operational use.

Human Evaluation

Experts review:

- High-risk predictions
- Low-risk predictions

and assess usefulness.

14. System Architecture

Ingestion Layer

Project records

Satellite imagery

Road network data

↓

Processing Layer

Geocoding

Cloud masking

Image extraction

↓

Feature Layer

Temporal features

Spectral features

Deep embeddings

↓

Model Layer

XGBoost

Siamese CNN

↓

Scoring Layer

Completion probability

↓

Dashboard Layer

Project ranking

Visual evidence

Risk score

15. Ethical Considerations

The system does not identify corruption.

The system does not accuse individuals.

The system only identifies discrepancies between:

Reported status

and

Observable physical evidence.

Human review remains essential.

False positives may arise due to:

- Vegetation cover
- Cloud contamination
- Construction techniques

- Temporal mismatch

Outputs should therefore be interpreted as audit leads rather than conclusions.

16. Risks

Risk 1

Poor project geolocation.

Mitigation:

Restrict dataset to projects with reliable coordinates.

Risk 2

Insufficient visible change.

Mitigation:

Select major road projects only.

Risk 3

Cloud cover.

Mitigation:

Multi-date compositing.

Risk 4

Small dataset.

Mitigation:

Transfer learning.

17. Expected Deliverables

1. Curated infrastructure monitoring dataset.
 2. Geospatial preprocessing pipeline.
 3. XGBoost baseline model.
 4. Siamese CNN model.
 5. Comparative evaluation report.
 6. Interactive project risk map.
 7. Ranked list of projects requiring review.
-

18. Expected Contributions

Technical:

- Application of deep learning to infrastructure monitoring.
- Integration of remote sensing and procurement analytics.

Policy:

- Independent infrastructure verification.
- Scalable public accountability tool.
- Early warning system for project delays.

Research:

- Novel intersection of public procurement and computer vision.
 - Practical framework for monitoring infrastructure delivery using satellite imagery.
-

19. Conclusion

This project investigates whether multi-temporal satellite imagery can provide an independent and scalable mechanism for verifying government-reported road infrastructure completion.

By combining public procurement records with deep learning-based change detection, the proposed system aims to identify projects whose reported status may warrant further human review.

The resulting framework demonstrates how modern geospatial AI can contribute to transparency, accountability, and evidence-based governance while remaining technically feasible within a short research and development timeline.